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By the end of this lesson, I will be able to:

- Find equivalent fractions with common denominators.
- Add fractions with different denominators.
- Add mixed numbers with different denominators.

1. Find Equivalent Fractions with Common Denominators

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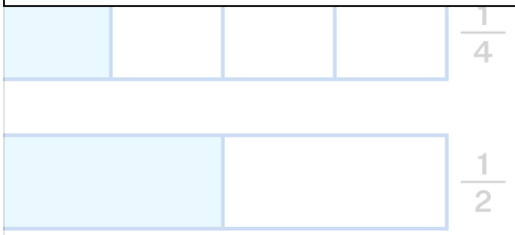

The top strip shows fourths.

The bottom strip shows halves.

To combine the fractions, the pieces must be the same size.

Each half can be split into 2 equal parts to make fourths.

Now both fractions use fourths.

What is the relationship between the denominators, 2 and 4?

- 2 is a factor of 4 ✔
- 4 and 2 have no common factors
- 4 is a factor of 2
- 4 and 2 are equal

1a

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So we used 4 as the common denominator.

> When one denominator is a **factor** of the other, use the **larger denominator** as the common denominator.
>
> To make equivalent fractions, multiply the numerator and denominator by the same number.

Now try another pair of fractions.

Ava has $\frac{2}{3}$ meter of orange ribbon.

Liam has $\frac{1}{9}$ meter of orange ribbon.

Since 3 is a factor of 9, use **9** as the common denominator.

Convert $\frac{2}{3}$ into an equivalent fraction with denominator 9.

$$\frac{2}{3} = \frac{2 \times \square}{3 \times \square} = \frac{\square}{9}$$

The fraction $\frac{1}{9}$ already has denominator 9, so it stays the same:

$$\frac{\square}{\square}$$

Now both fractions are written in ninths.

Are both ribbons measured using the same-sized parts?

Type '1' for Yes and '0' for No. $\square$

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1b

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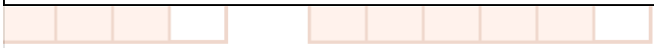

$\frac{3}{4}$ $\frac{5}{6}$

Can we use fourths or sixths as the common denominator?

- $6 \div 4$ does not divide evenly.
- $4 \div 6$ does not divide evenly.

So we need a new common denominator.

1c

List multiples of 4 and 6.

- Multiples of 4: $4, 8, 12, 16, 20 \ldots$
- Multiples of 6: $6, 12, 18, 24 \ldots$

The smallest common multiple is [ ].

So use [ ] as the common denominator.

Correct Answers:

1 12

2 12

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The **smallest common multiple** becomes the common denominator.

We saw that the smallest common multiple of 4 and 6 is 12, so the fraction strips of fourths and sixths could be divided into 12 equal strips.

Let us express $\frac{1}{4}$ and $\frac{1}{6}$ with the common denominator 12.

$$\frac{3}{4} = \frac{3 \times \square}{4 \times \square} = \frac{\square}{12}$$

1d

$$\frac{5}{6} = \frac{5 \times \square}{6 \times \square} = \frac{\square}{12}$$

Both fractions now have denominator $\square$.

Can these fractions now be combined easily? Type 1 for 'Yes' and 0 for 'No'. $\square$

Correct Answers:

1 $\frac{3}{4} = \frac{3\times3}{4\times3} = \frac{9}{12}$

2 $\frac{5}{6} = \frac{5\times2}{6\times2} = \frac{10}{12}$

3 12

4 1

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List multiples to find the common denominator:

- Multiples of 8: $8, 16, 24, \square, 40 \ldots$
- Multiples of 12: $12, \square, 36, 48 \ldots$

The smallest common multiple is $\square$.

Now write equivalent fractions with the common denominator.

$$\frac{5}{8} = \frac{5 \times \square}{8 \times \square} = \frac{\square}{24}$$

1e

$$\frac{3}{12} = \frac{3 \times \square}{12 \times \square} = \frac{\square}{24}$$

Correct Answers:

1. 32
2. 24
3. 24
4. $\frac{5}{8} = \frac{5 \times 3}{8 \times 3} = \frac{15}{24}$
5. $\frac{3}{12} = \frac{3 \times 2}{12 \times 2} = \frac{6}{24}$

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First, check for common factors between 3 and 5:

Number	Factors
3	1, 3
5	1, 5

The only common factor is 1.

When two denominators share no common factors except 1, let's see what happens when we list their multiples:

Find Multiples	List
Multiples of 3	$3, 6, 9, 12, 15, 18 \ldots$
Multiples of 5	$5, 10, 15, 20, 25 \ldots$

The first common multiple is 15.

Notice that $15 = 3 \times 5 = 15$.

Now we write equivalent fractions to convert the fractions to have a common denominator 15:

$$\frac{1}{3} = \frac{1 \times \square}{3 \times \square} = \frac{\square}{15}$$

$$\frac{1}{5} = \frac{1 \times \square}{5 \times \square} = \frac{\square}{15}$$

Correct Answers:

1 $\frac{1}{3} = \frac{1 \times 5}{3 \times 5} = \frac{5}{15}$

2 $\frac{1}{5} = \frac{1 \times 3}{5 \times 3} = \frac{3}{15}$

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common denominator.

Find equivalent fractions with a common denominator for $\frac{4}{7}$ and $\frac{5}{9}$.

The denominators 7 and 9 share no common factors except 1.

So:

$7 \times 9 = 63$

Use 63 as the common denominator.

Write the equivalent fractions with the common denominator 63.

$\frac{4}{7} = \frac{\square}{\square}$

$\frac{5}{9} = \frac{\square}{\square}$

Correct Answers:

1 $\frac{36}{63}$

2 $\frac{35}{63}$

1g

2. Add Fractions with Different Denominators

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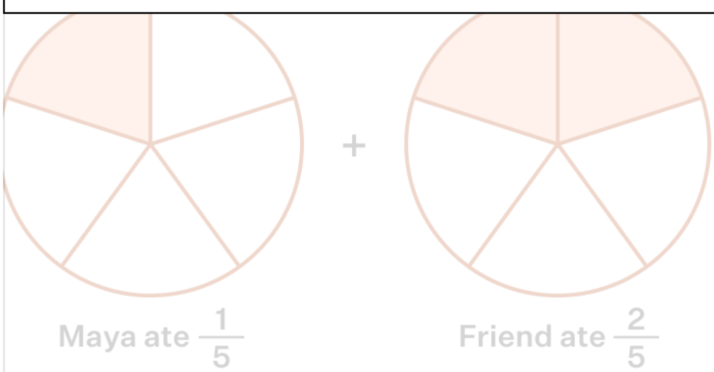

Maya eats $\frac{1}{5}$ of the pizza and her friend eats $\frac{2}{5}$ of the pizza.

Since both amounts use fifths, we can add the parts directly:

$\frac{1}{5} + \frac{2}{5} = \frac{\square}{5}$

Together they ate $\frac{\square}{\square}$ of the pizza.

Correct Answers:

1 $\frac{1}{5} + \frac{2}{5} = \frac{3}{5}$

2 $\frac{3}{5}$ of the pizza.

2a

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Click Play!

So,

To add fractions with different denominators, we must first make the parts the same size.

$\frac{1}{2} + \frac{1}{3} = \frac{\square}{6} + \frac{\square}{6} = \frac{\square}{\square}$

Correct Answers:

1

$\frac{1}{2} + \frac{1}{3} = \frac{3}{6} + \frac{2}{6} = \frac{5}{6}$

2b

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To add fractions with different denominators:

1. Convert both fractions to equivalent fractions with the common denominator

2. Add the numerators and keep the common denominator

Let's add $\frac{1}{2} + \frac{1}{4}$ using these steps.

Step	Work
Step 1: Find common denominator and convert	Since 2 is a factor of 4, use 4 as the common denominator $\frac{1}{2} = \frac{1 \times \square}{2 \times \square} = \frac{\square}{4}$ $\frac{1}{4}$ already has denominator 4
Step 2: Add	$\frac{\square}{4} + \frac{1}{4} = \frac{\square}{\square}$

Correct Answers:

1. $\frac{1}{2} = \frac{1 \times 2}{2 \times 2} = \frac{2}{4}$

2. $\frac{2}{4} + \frac{1}{4} = \frac{3}{4}$

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2c

2d

Sam is making a banner and needs $\frac{1}{4}$ meter of blue fabric and $\frac{2}{6}$ meter of red fabric. Let's find the total length.

Add $\frac{1}{4} + \frac{2}{6}$

Step	Work
Step 1: Find common denominator and convert fractions	The smallest common multiple of 4 and 6 is 12. $\frac{1}{4} = \frac{1 \times 3}{4 \times 3} = \frac{3}{12}$ $\frac{2}{6} = \frac{2 \times \square}{6 \times \square} = \frac{\square}{12}$
Step 2: Add	$\frac{3}{12} + \frac{\square}{12} = \frac{\square}{\square}$

Correct Answers:

1 $\frac{2}{6} = \frac{2\times2}{6\times2} = \frac{4}{12}$

2 $\frac{3}{12} + \frac{4}{12} = \frac{7}{12}$

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2e

Alex uses $\frac{1}{3}$ liter of orange juice and $\frac{2}{5}$ liter of apple juice for punch. Find the total amount of juice in liters.

Add $\frac{1}{3} + \frac{2}{5}$

Step	Work
Step 1: Find common denominator and convert fractions	3 and 5 share no common factors except 1 Common denominator $= 3 \times 5 = 15$ $\frac{1}{3} = \frac{1 \times 5}{3 \times 5} = \frac{5}{15}$ $\frac{2}{5} = \frac{2 \times \square}{5 \times \square} = \frac{\square}{15}$
Step 2: Add	$\frac{5}{15} + \frac{\square}{15} = \frac{\square}{\square}$
Step 3: Answer	Total amount $= \frac{\square}{\square}$ liters

Correct Answers:

1. $\frac{2}{5} = \frac{2 \times 3}{5 \times 3} = \frac{6}{15}$
2. $\frac{5}{15} + \frac{6}{15} = \frac{11}{15}$
3. $\frac{11}{15}$ liters

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